

ECON 100B: Macroeconomics

Class 8: The Solow Model 3 (Jones Chapter 5)

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February 12, 2026

Agenda

- Announcements
- Current events
- Chapter 5: The Solow Growth Model



BERKELEY STUDENT LEARNING CENTER

ECONOMICS PROGRAM

Mastery through Collaborative Learning

ECON 100B

César Chávez Student Center

Monday - Thursday 12 - 5 PM



Service Schedule

Announcements (same from Class 05)

- Problem Set 1 is due on Gradescope on Friday 2/13
- Problem Set 2 will be due on Gradescope on Friday 2/27
- In-class Midterm the next Thursday 3/5
- Take-home Midterm Essay released 3/8, due 3/20
- Spring Break 3/23-3/27

Read both of these newspapers

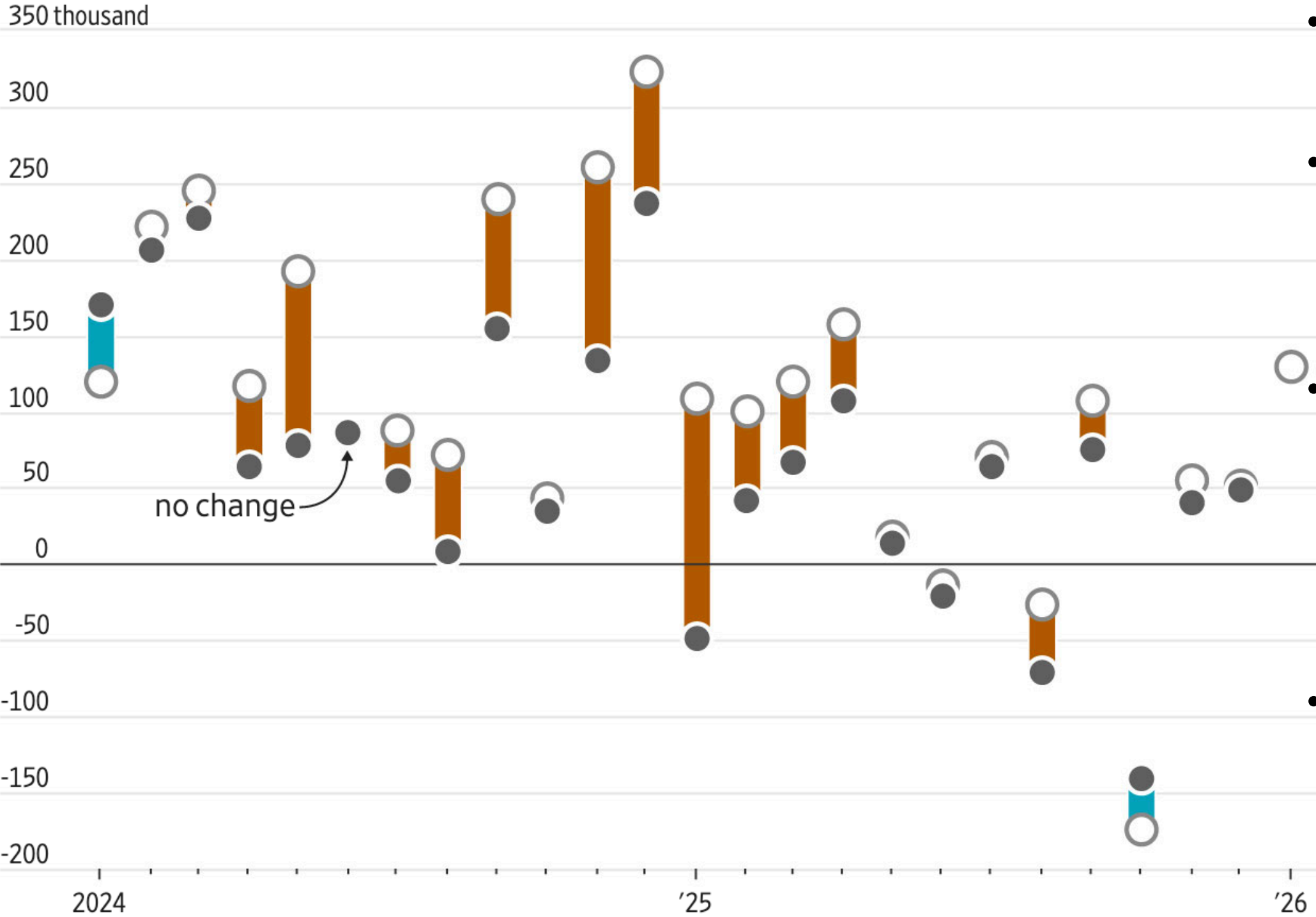
- Wall Street Journal
- New York Times
- For access to both, see [this Library guide](#)
- You might skip the op-ed pages unless you like being horrified or vindicated, depending on your point of view
- But sometimes there are very good op-ed pieces in both

Current events

- [January 2026 Employment Situation](#) released on Wed Feb 11
- Like a lot of recent macroeconomic news, it's mixed
- Big downward revisions in payroll employment in 2024
- But a big +130,000 increase in January
 - Driven by health care [per WSJ](#) and others
- Unemployment rate at 4.3%, down from 4.4%
 - [Justin Wolfers \(Michigan, Brookings\) points out](#) that the survey response rate was lower than usual because of the winter cold
 - Lower response = more uncertainty in the true population mean

Nonfarms payrolls, change from a month earlier

● Revised ○ Previous ■ Decline ■ Growth



- This graphic from [Chen, WSJ](#)

- [Lahart, WSJ](#) discusses some reasons for the changes:

- “Participation bias” with firms that answered the survey having stronger job growth

- Firm “birth-death” model revisions.

You can’t survey brand-new firms

Note: Seasonally adjusted
Source: Labor Department

Under the radar in several ways

- Grossman, WSJ (1/30/26) “[Trump Picks Veteran Staffer to Head Bureau of Labor Statistics](#)”
- Brett Matsumoto, economics Ph.D., UNC 2015
- Last August 1, after BLS data revisions, Trump had fired the BLS commissioner and claimed someone was rigging the numbers
- Later in August, Trump had nominated the chief economist at the Heritage Foundation, E.J. Antoni (econ Ph.D. NIU 2020)
- Before the Senate could vote no, the administration pulled the nomination in September

First half of ECON 100B

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
 - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y]$$

$$g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

- Chapter 4 introduces Cobb-Douglas production

$$Y = A K^{1/3} L^{2/3}$$

with constant returns to scale
and declining marginal productivity of K or L

- **Chapter 5 develops the Solow Model**
- Chapter 6 develops the Romer Model

Learning objectives in Chapter 5

- Investment in new capital comes from **saving**, which usually comes from domestic income not consumed: $S = Y - C = I$
- If people work and save at rate s out of income, then $S = sY$
- But capital also breaks at rate d
- The Solow Model includes 2 curves: **saving** sY , and **depreciation** dK ,
and where they cross, $sY = dK$, capital is unchanging at K^*
in what we call a **steady state**
- When countries are far below their steady states, they **grow fast**
- A simple extension in §5.10 is a steady state with $k^* = \frac{K}{L}$ with
population growth at rate n

The Solow model with population growth in §5.10

- When saving becomes investment, and depreciation destroys K at rate d , then

$$\Delta K = sY - dK$$

- It turns out that when L is growing at rate n , this also implies that

$$\Delta k = sy - (n + d)k$$

$$\text{when } k \equiv \frac{K}{L} \text{ and } y \equiv \frac{Y}{L}$$

- Surprise! The Solow model with population growth **looks almost exactly the same**

The math

- Start by taking growth rates of both sides of $k \equiv \frac{K}{L}$

$$g[k] = g[K] - g[L]$$

- Growth rates are the change over the level:

$$g[k] = \frac{\Delta k}{k} = \frac{\Delta K}{K} - \frac{\Delta L}{L} = g[K] - g[L]$$

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- But we know that $\Delta K = sY - dK$, and we also know $g[L] = n$

$$\frac{\Delta k}{k} = \frac{sY - dK}{K} - n$$

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$$\frac{\Delta k}{k} = \frac{sY - dK}{K} - n$$

$$\frac{\Delta k}{k} = \frac{sY}{K} - \frac{dK}{K} - n$$

$$\frac{\Delta k}{k} = \frac{sY}{K} - d - n$$

The finish line

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$$\frac{\Delta k}{k} = \frac{sY}{K} - d - n$$

- Now use $Y = yL$ and $K = kL$ because $y \equiv \frac{Y}{L}$ and $k \equiv \frac{K}{L}$

$$\frac{\Delta k}{k} = \frac{syL}{kL} - d - n$$

The finish line

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- Now use $Y = yL$ and $K = kL$ because $y \equiv \frac{Y}{L}$ and $k \equiv \frac{K}{L}$

$$\frac{\Delta k}{k} = \frac{syL}{kL} - d - n$$

now simplify

$$\frac{\Delta k}{k} = \frac{sy}{k} - (n + d)$$

- Multiply through by k

$$\Delta k = sy - (n + d)k$$

Where we are

- Solow model with population growth:

$$y = \bar{A}k^{1/3}$$

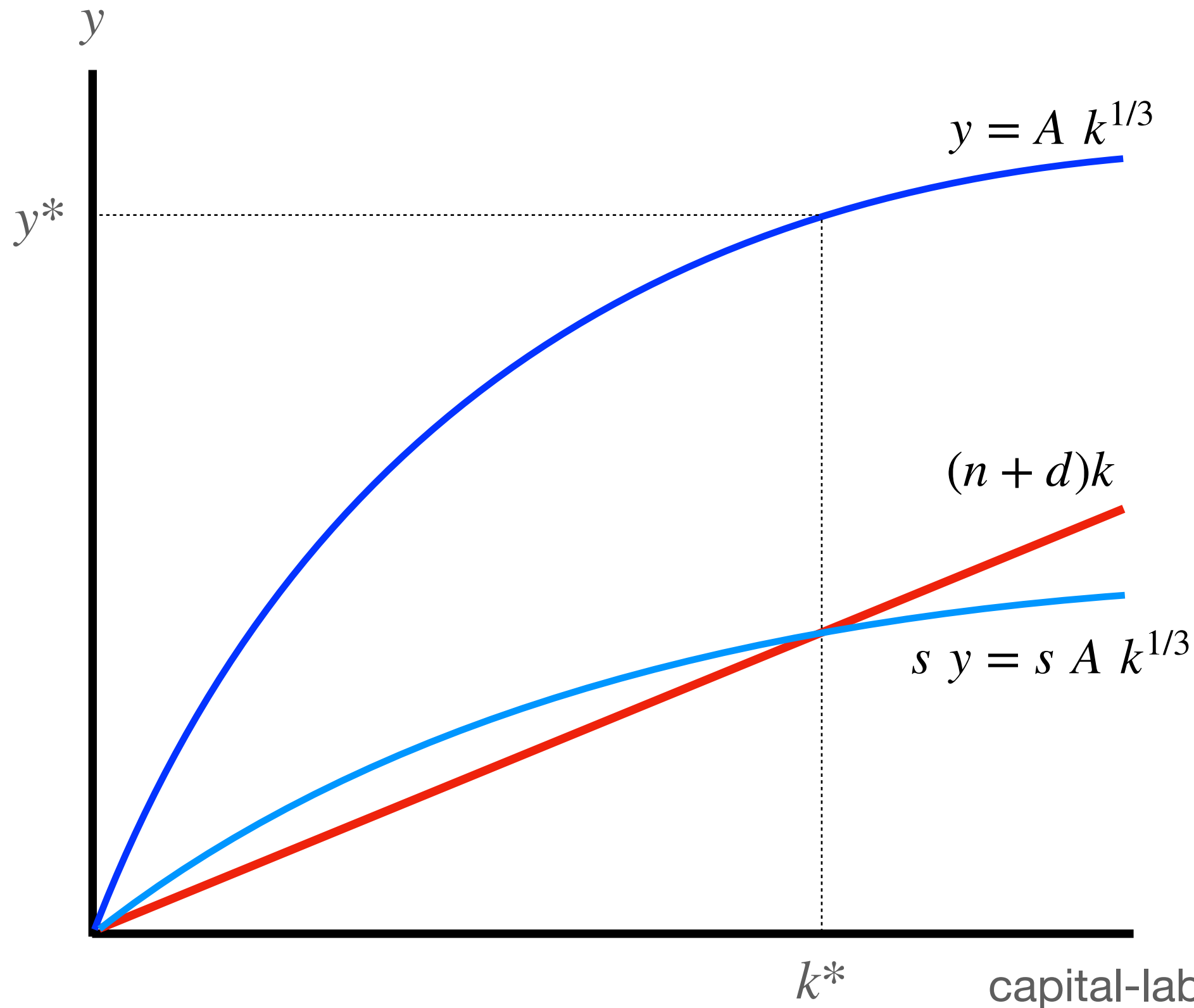
output per worker $y \equiv \frac{Y}{L}$

$$\Delta k = sy - (n + d)k$$

the change in capital per worker is saving per worker minus total depreciation per worker

- Total depreciation is the population growth rate plus the depreciation rate
- The intuition: Capital per worker falls either when capital breaks or when the labor supply grows

The Solow diagram shows output per worker as a function of capital per worker



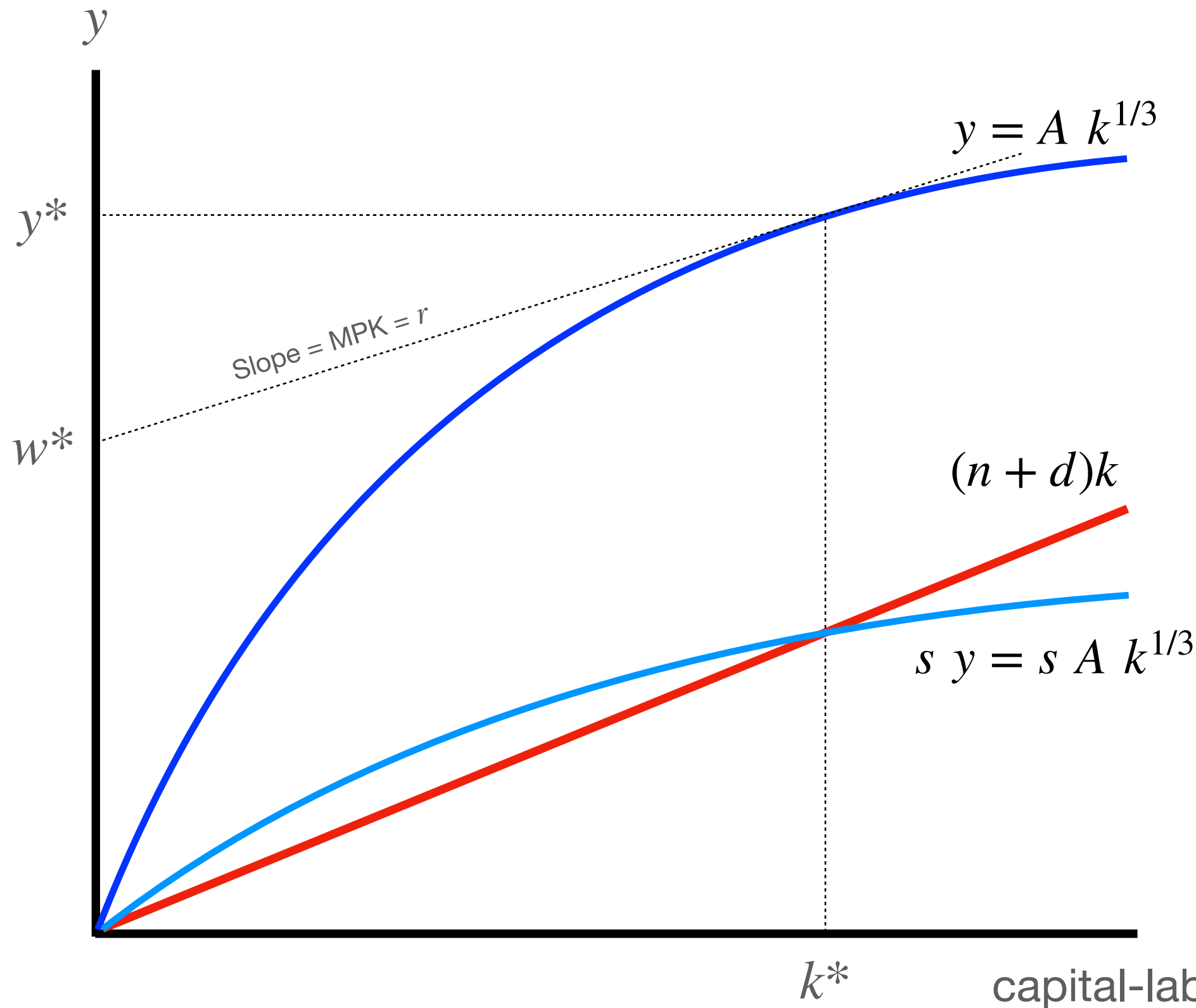
- Because capital can adjust, the economy will return to k^* unless there are changes to the parameters n , d , A , s , or $\alpha = 1/3$

- It turns out that we can solve for k^*

More soon

$$k^* = \left(\frac{s A}{n + d} \right)^{3/2}$$

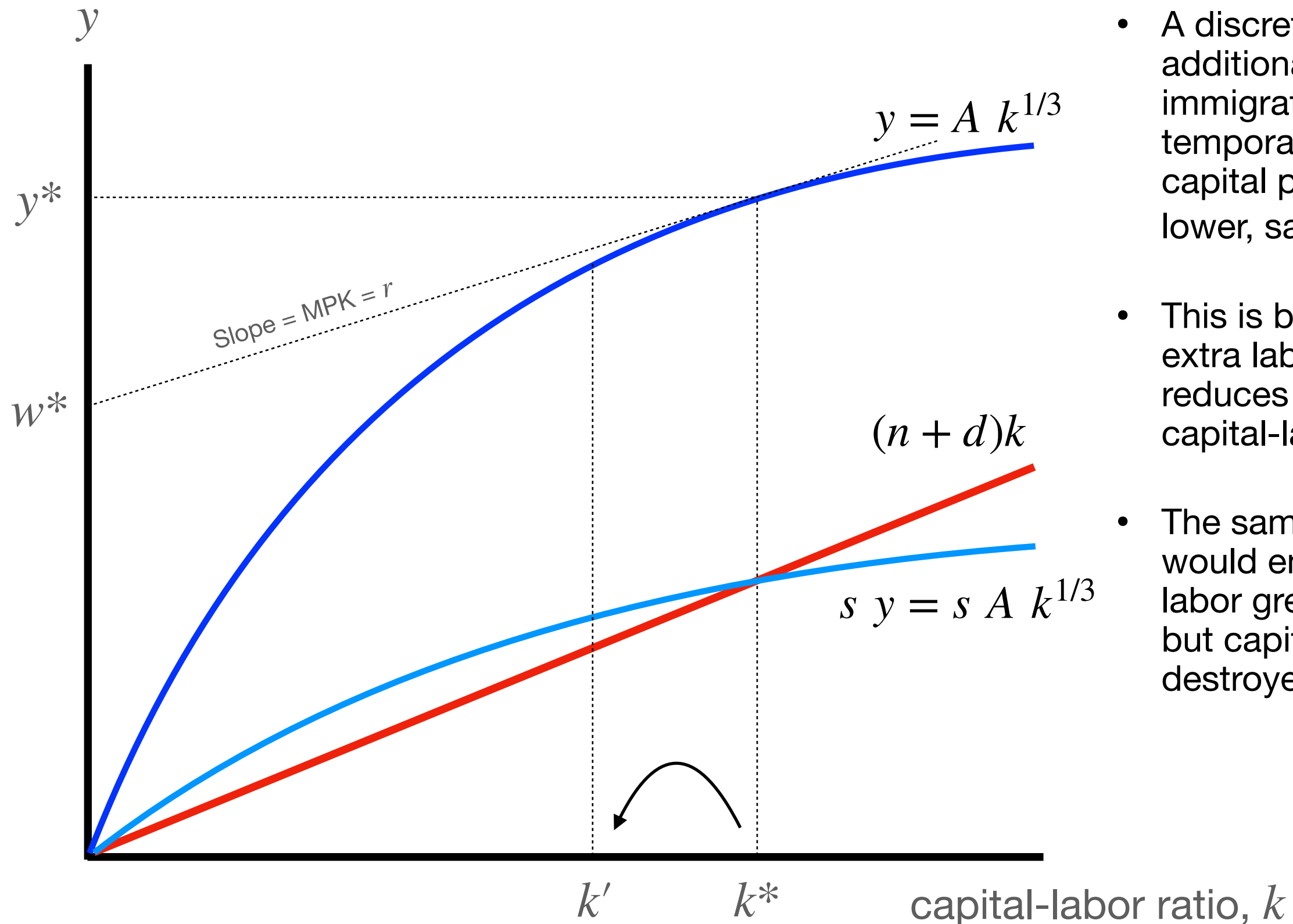
It turns out that we can also see the returns to capital and the wage



- The tangent line to y has slope equal to r , the marginal product of capital a.k.a. the interest rate
- Slope r must be rise over run
- Run is k^*
- Rise must be $r k^*$
- Income per worker y^* is the wage per worker plus capital income per worker

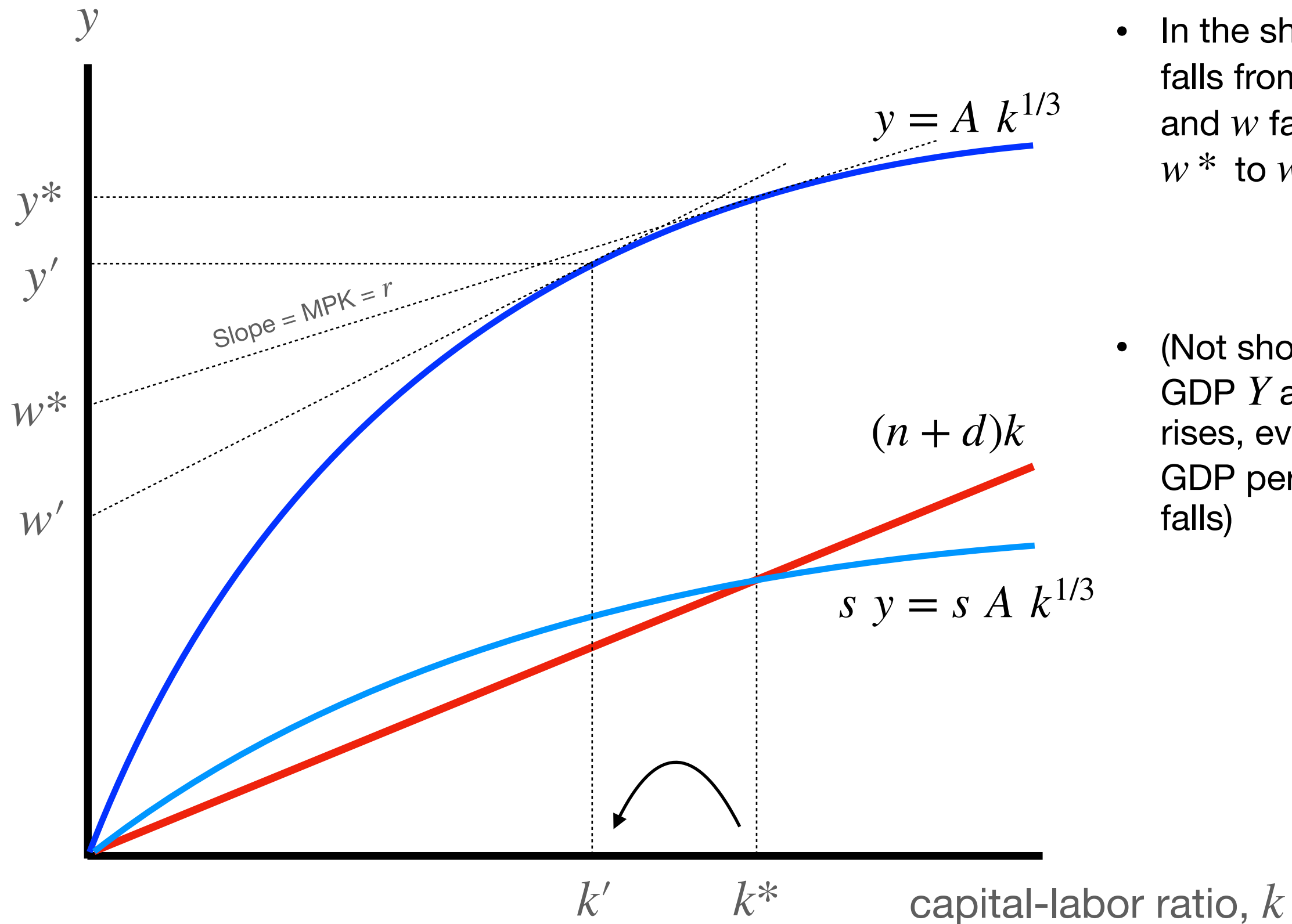
$$y^* = w + r k^*$$

In the Solow model with population growth, shocks to labor will only produce temporary changes in the wage



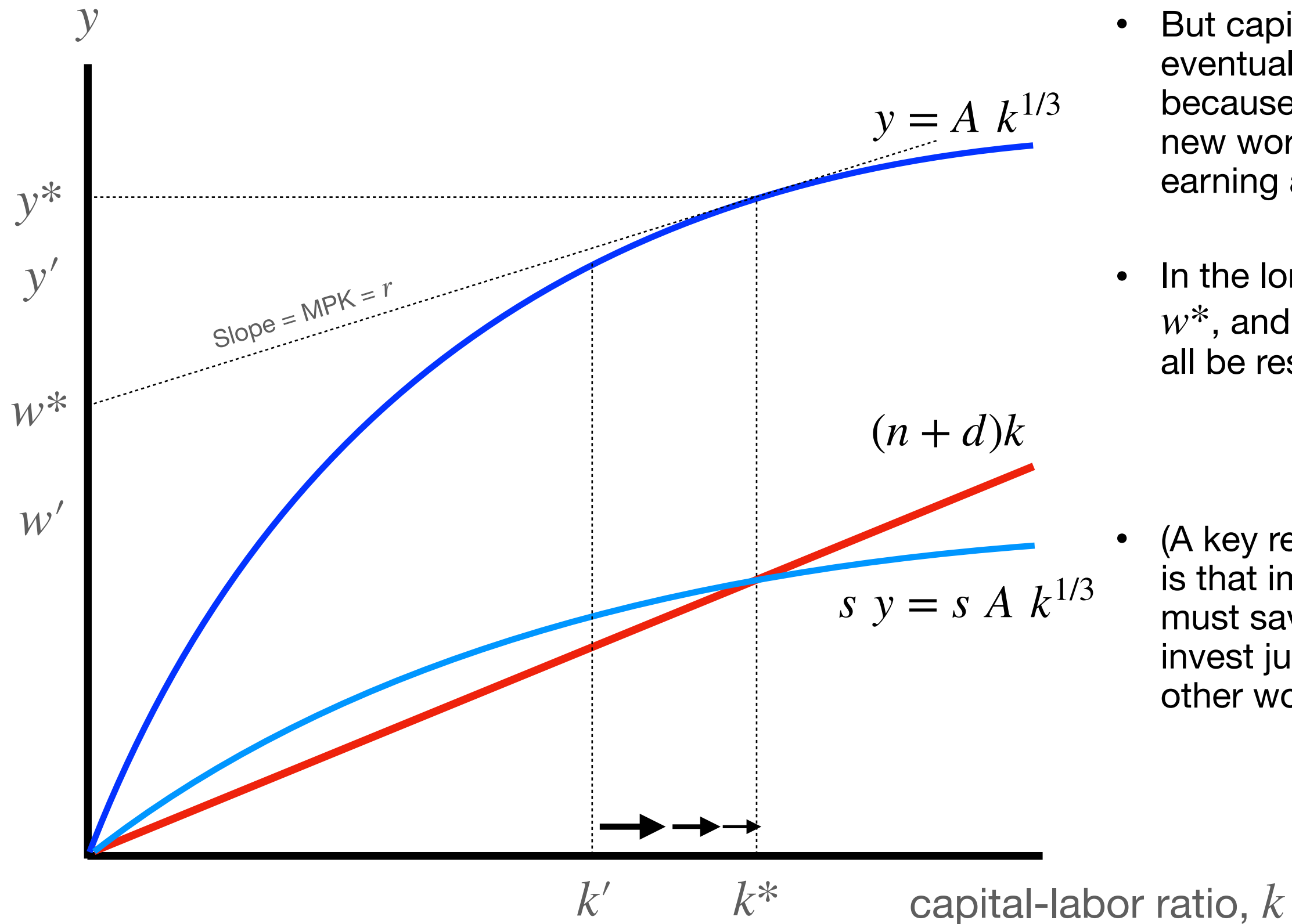
- A discrete wave of additional immigration would temporarily push capital per worker lower, say to k'
- This is because extra labor reduces the capital-labor ratio
- The same picture would emerge if labor grew steadily but capital were destroyed

Short-run effects of a one-time wave of immigration



- In the short run, y falls from y^* to y' and w falls from w^* to w'
- (Not shown here: GDP Y actually rises, even though GDP per worker y falls)

Long-run effects of a one-time wave of immigration?



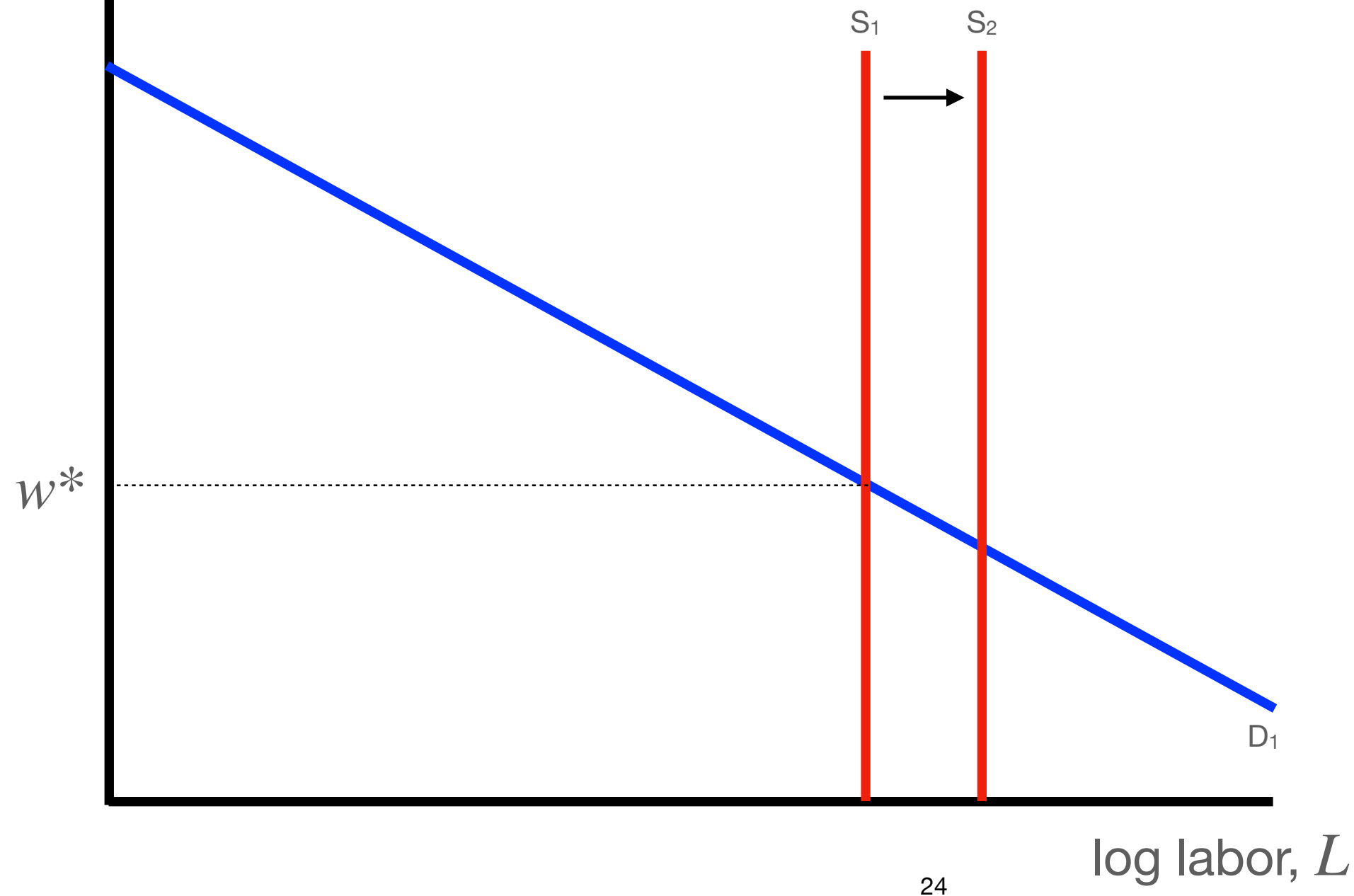
- But capital would eventually adjust because those new workers are earning and saving
- In the long run, k^* , w^* , and y^* would all be restored
- (A key requirement is that immigrants must save and invest just like other workers)

When capital adjusts, the labor demand curve shifts

log wage, w

$$w = \frac{2}{3} A \left(\frac{K}{L} \right)^{1/3}$$

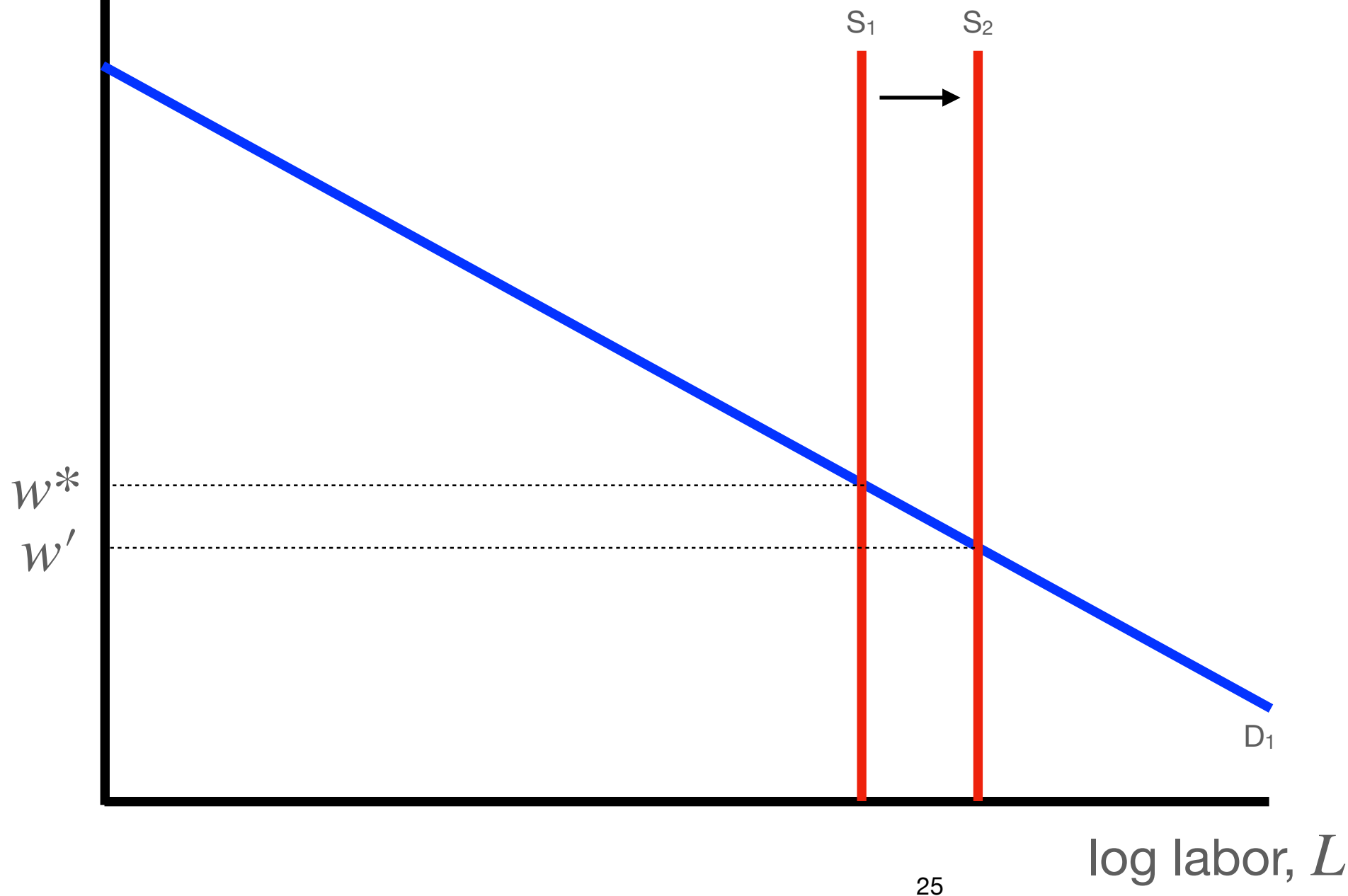
- Suppose immigration bumps labor higher



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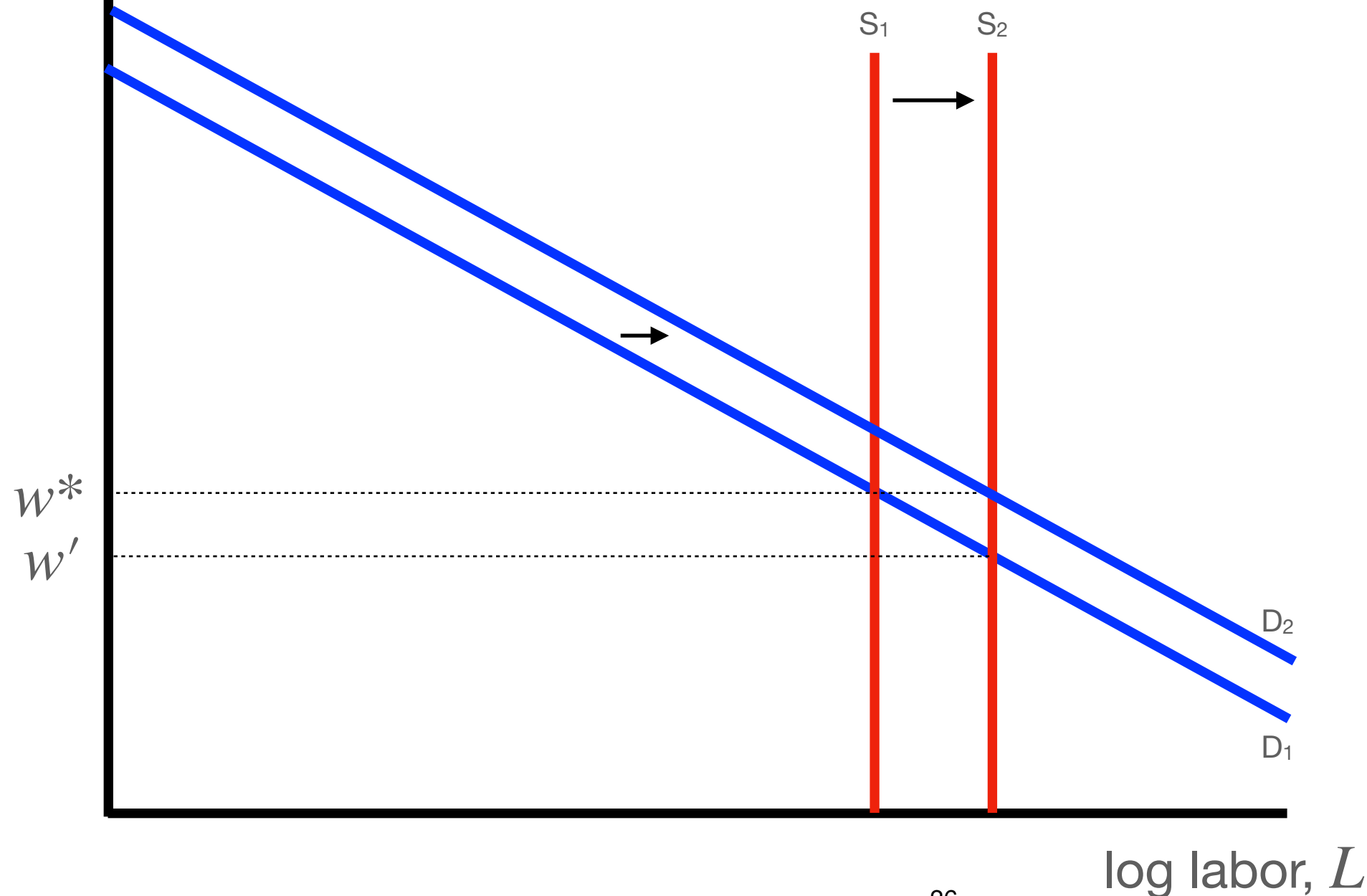


- Suppose immigration bumps labor higher
- The wage falls in the short run

When capital adjusts, the labor demand curve shifts

log wage, w

$$w = \frac{2}{3} A \left(\frac{K}{L} \right)^{1/3}$$



- Suppose immigration bumps labor higher
- The wage falls in the short run
- But if capital adjusts like it does in the Solow model, labor demand will shift out, restoring w^* eventually

Punchlines

- Capital adjusts
- Shocks to the levels of labor or capital have short-term impacts
- Unless these shocks change the parameters n , d , A , s , or $\alpha = 1/3$, they will not change the steady-state income per worker or wage
- Does immigration change any of the parameters? Discuss.

Solving for the steady state

$$\Delta k = sy - (n + d)k$$

write out y

$$\Delta k = sAk^{1/3} - (n + d)k$$

When the economy is in steady state, $\Delta k = 0$, so therefore

$$\Delta k = sAk^{1/3} - (n + d)k = 0$$

$$sAk^{1/3} = (n + d)k$$

Collect terms

$$\frac{sA}{n + d} = \frac{k}{k^{1/3}} \quad \rightarrow \quad \frac{sA}{n + d} = k^{2/3}$$

Raise to 3/2

$$k^* = \left(\frac{sA}{n + d} \right)^{3/2}$$

Steady-state output

$$k^* = \left(\frac{sA}{n+d} \right)^{3/2}$$

$$y^* = Ak^{*1/3} \quad \rightarrow \quad y^* = A \left[\left(\frac{sA}{n+d} \right)^{3/2} \right]^{1/3}$$

Simplify

$$y^* = A \cdot A^{3/2 \cdot 1/3} \left[\left(\frac{s}{n+d} \right)^{3/2} \right]^{1/3}$$

$$y^* = A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}$$

The capital-output ratio

$$k^* = \left(\frac{sA}{n+d} \right)^{3/2} \qquad y^* = A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}$$

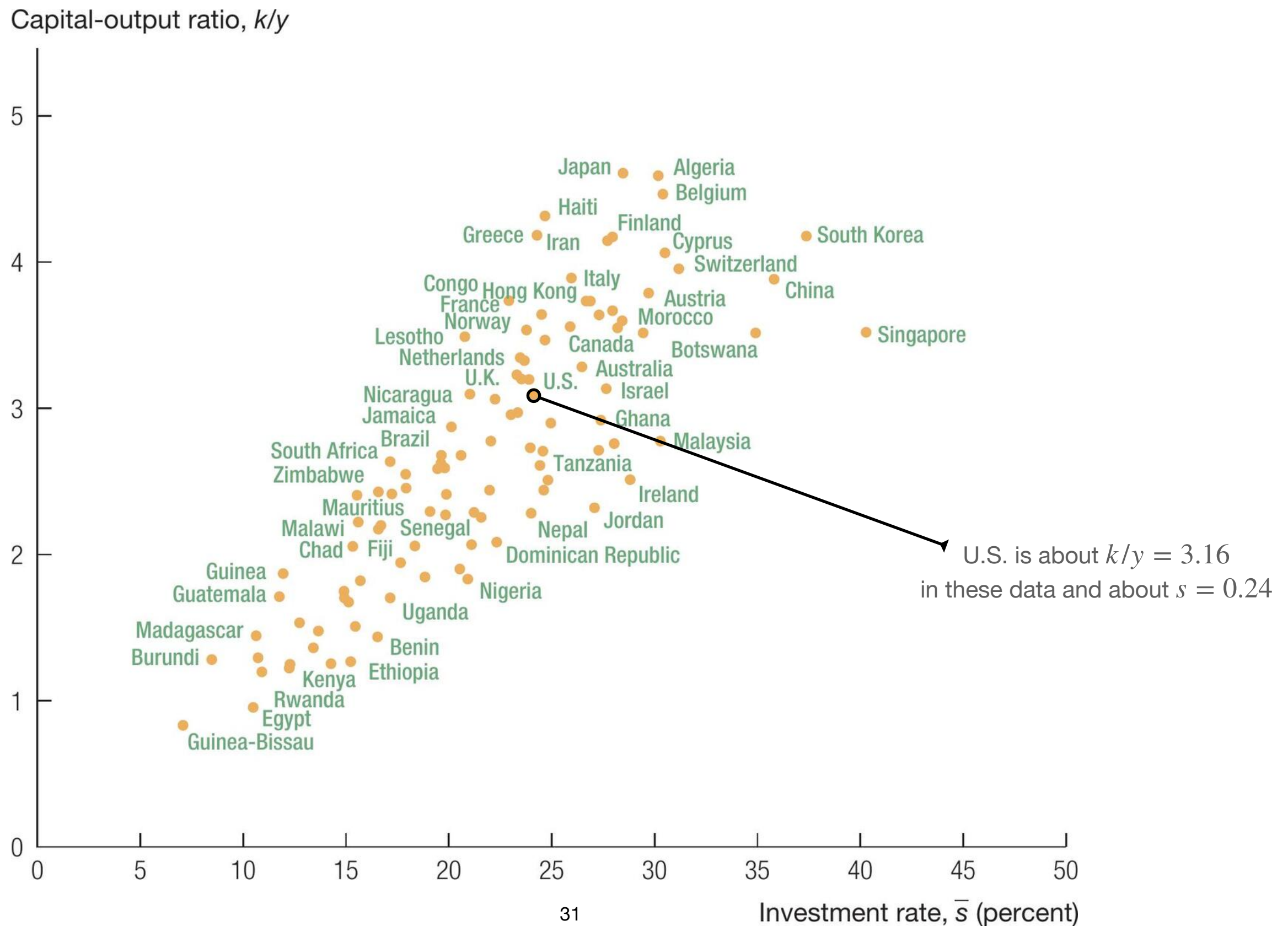
- Note that $\frac{K}{Y} = \frac{k}{y}$ because $k = \frac{K}{L}$ and $y = \frac{Y}{L}$

$$\frac{k^*}{y^*} = \frac{\left(\frac{sA}{n+d} \right)^{3/2}}{A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}} \qquad \frac{k^*}{y^*} = \frac{A^{3/2} \left(\frac{s}{n+d} \right)^{3/2}}{A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}}$$

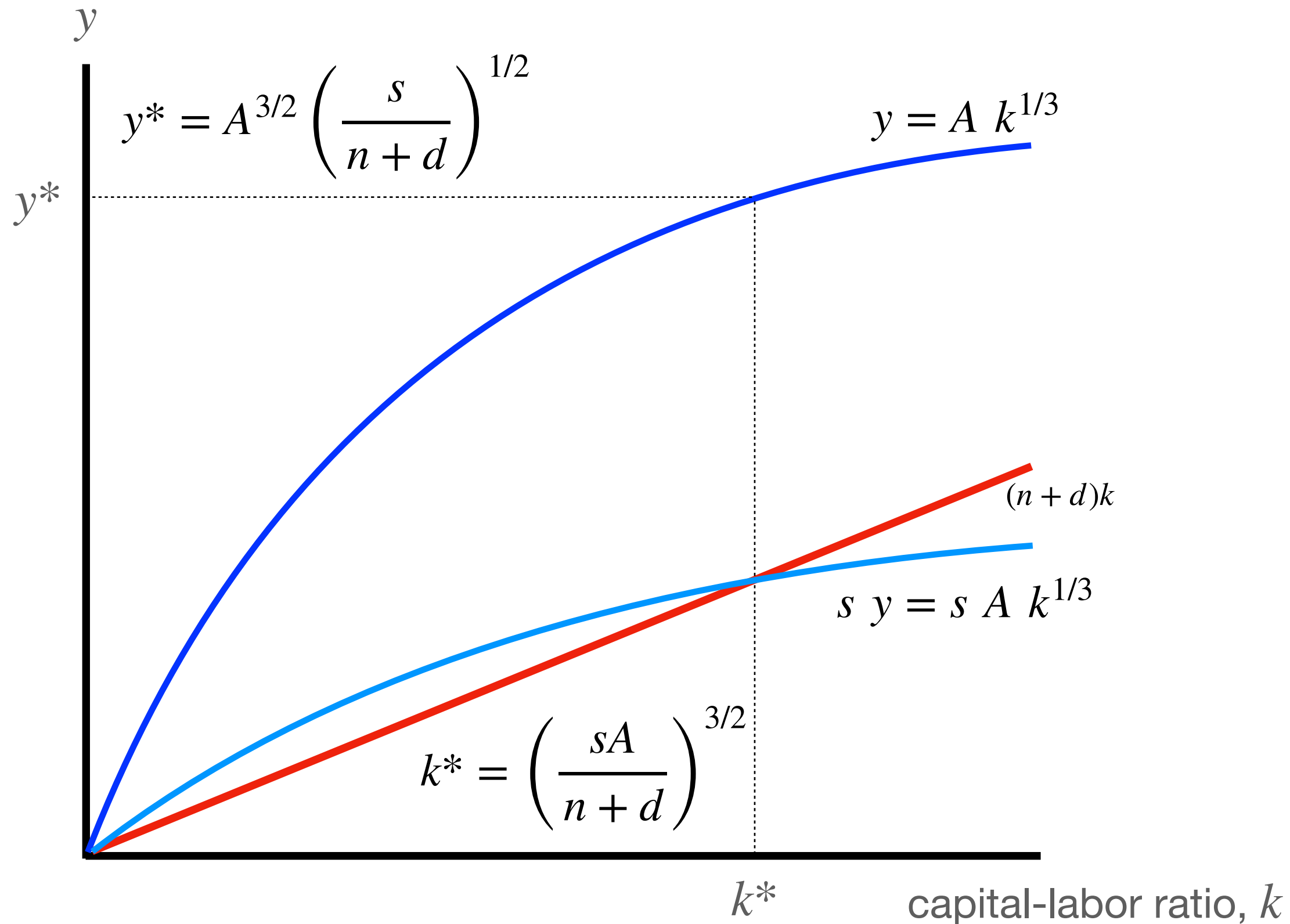
$$\frac{k^*}{y^*} = \frac{s}{n+d}$$

$$\frac{k^*}{y^*} = \frac{0.24}{0.01 + 0.05} = 4$$

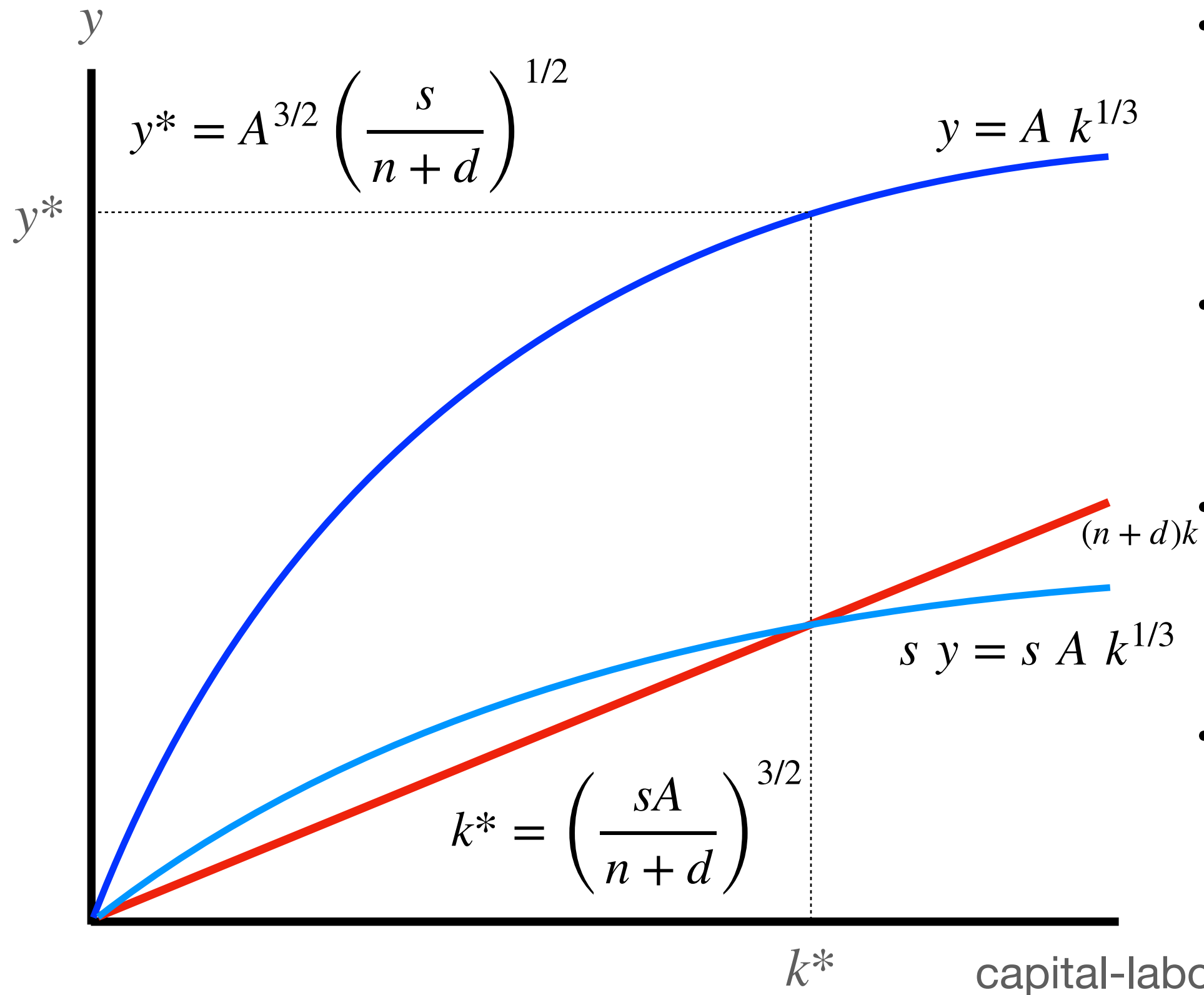
Figure 5.3 in §5.5 shows $\frac{k}{y}$ as a function of s across countries



Back to the diagram. What happens when population growth n increases?

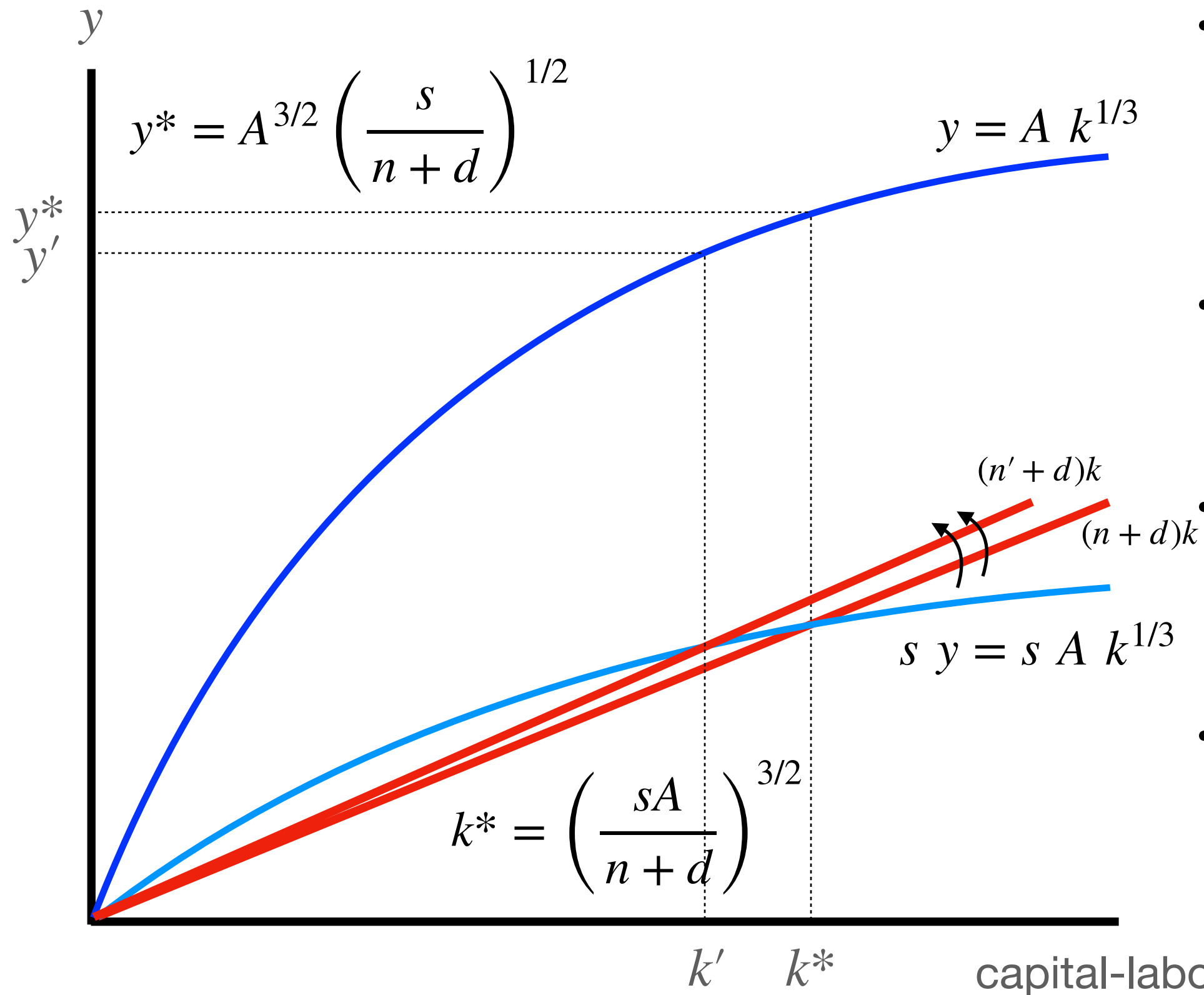


Population growth is a “bad thing” in the Solow model because it dilutes capital



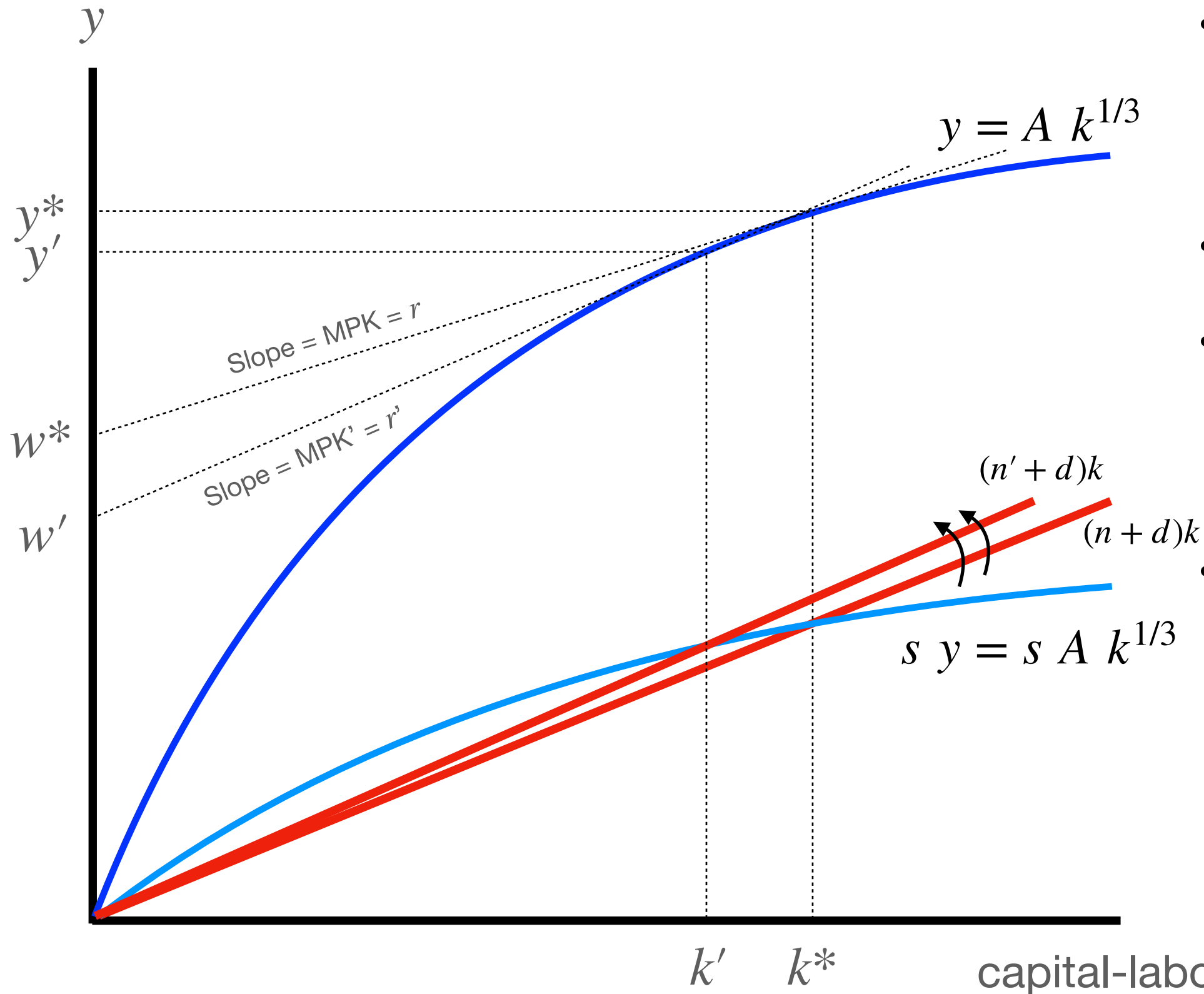
- By the math, the steady-state levels of capital and output will fall if n increases
- Suppose $n = 0.01$ and $d = 0.05$, so that $n + d = 0.06$
- If $n' = 0.016$ then $n' + d = 0.066$ or a 10% increase
- All else equal, that will:
reduce k^* by 15%
reduce y^* by 5%

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Faster population growth also reduces the wage. It increases the marginal product of capital



- In the Solow model, agents aren't really reacting to prices
- But if they were:
- A falling wage is a disincentive to immigrate and maybe to birth
- A rising $MPK = r$ represents increasing returns to capital and probably increases the demand for capital and saving

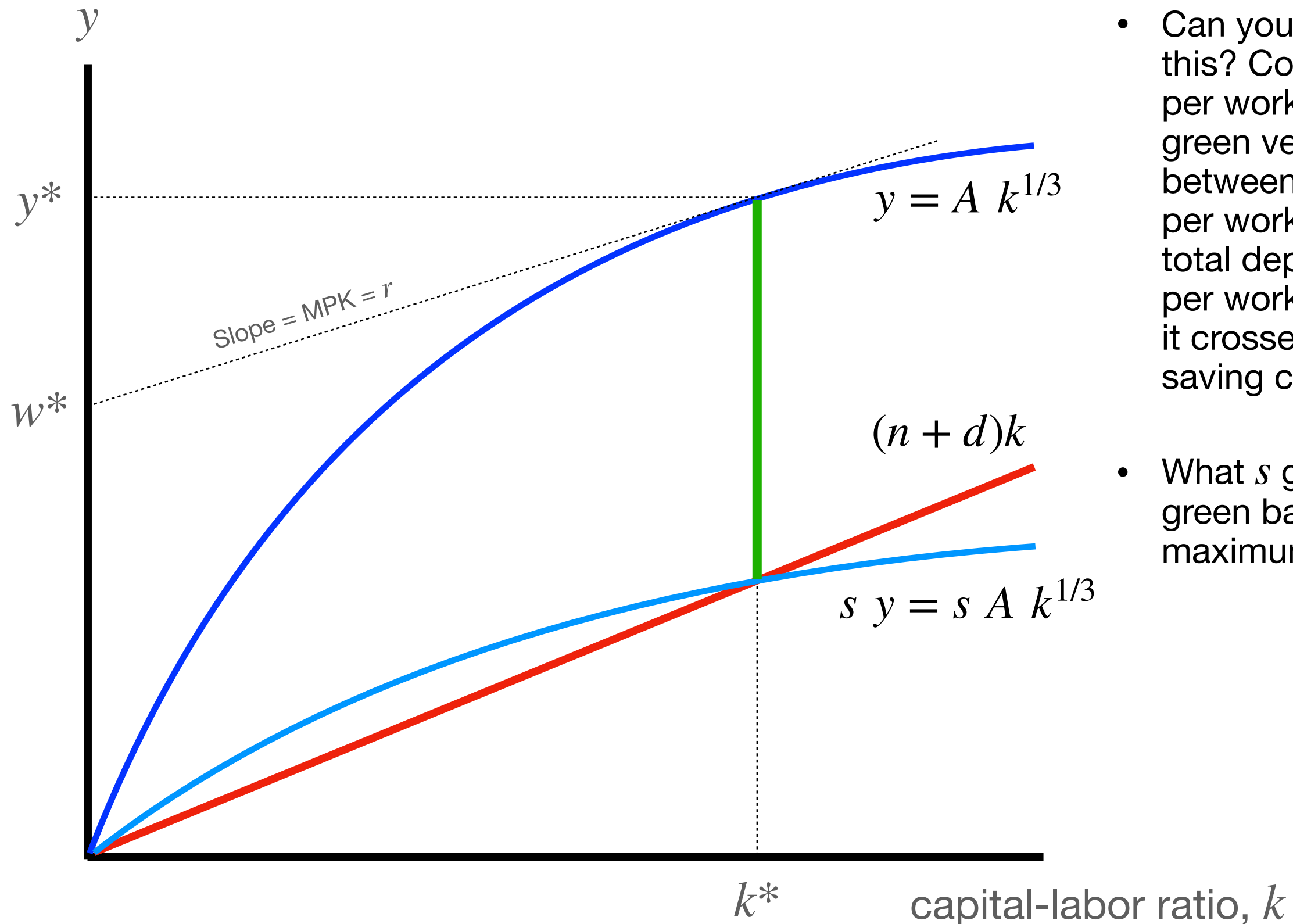
“Golden Rule” in the Solow model

Treat others as you would wish to be treated yourself

Time, generations, and sacrifice in the Solow model

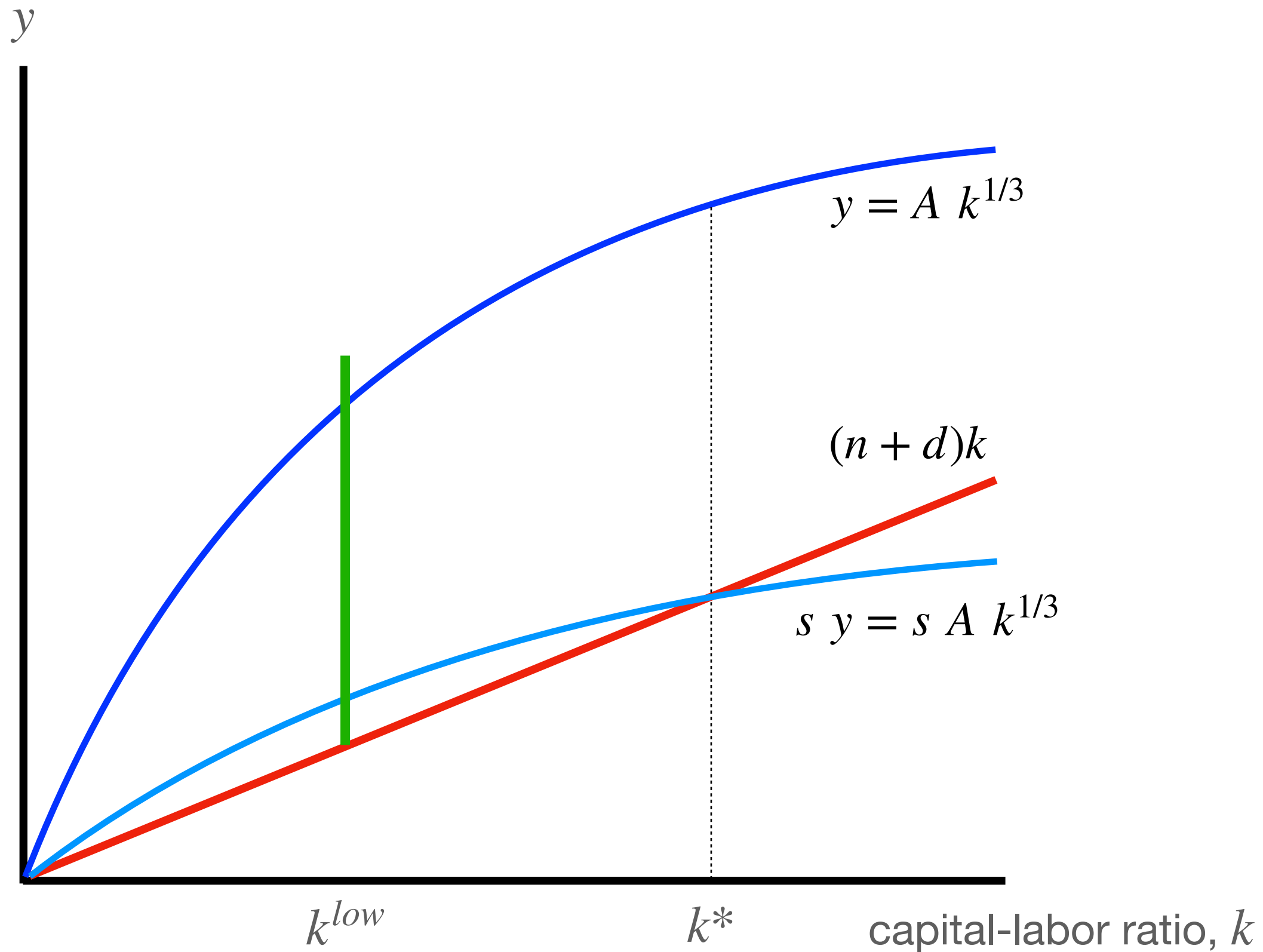
- Time passes in the Solow model, and real economic agents have finite lifetimes
- The capital stock K is wealth owned by the living and passed along to future generations
- We assume consumption $C = (1 - s)Y$ is based on income. Old people in the model consume out of income, not out of capital
- Utility depends on consumption. Capital is a means to the end of consumption today and in the future
- But we could maximize consumption today by setting $s = 0$ and swindling future generations
- What is the “right” rate of saving now?

What rate of saving maximizes consumption now and into the future?

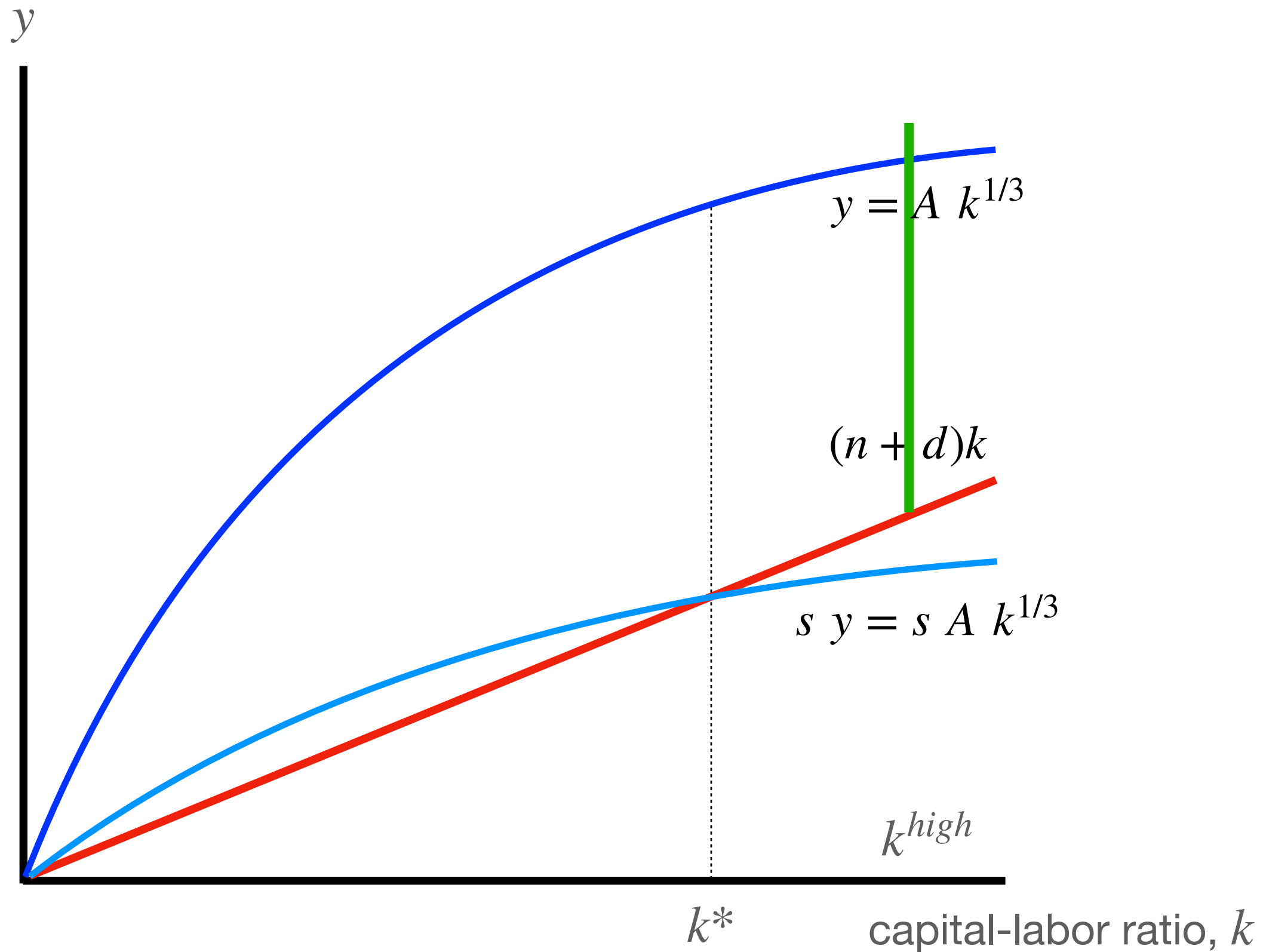


- Can you eyeball this? Consumption per worker is the green vertical bar between income per worker and total depreciation per worker, where it crosses the saving curve
- What s gets us the green bar with maximum length?

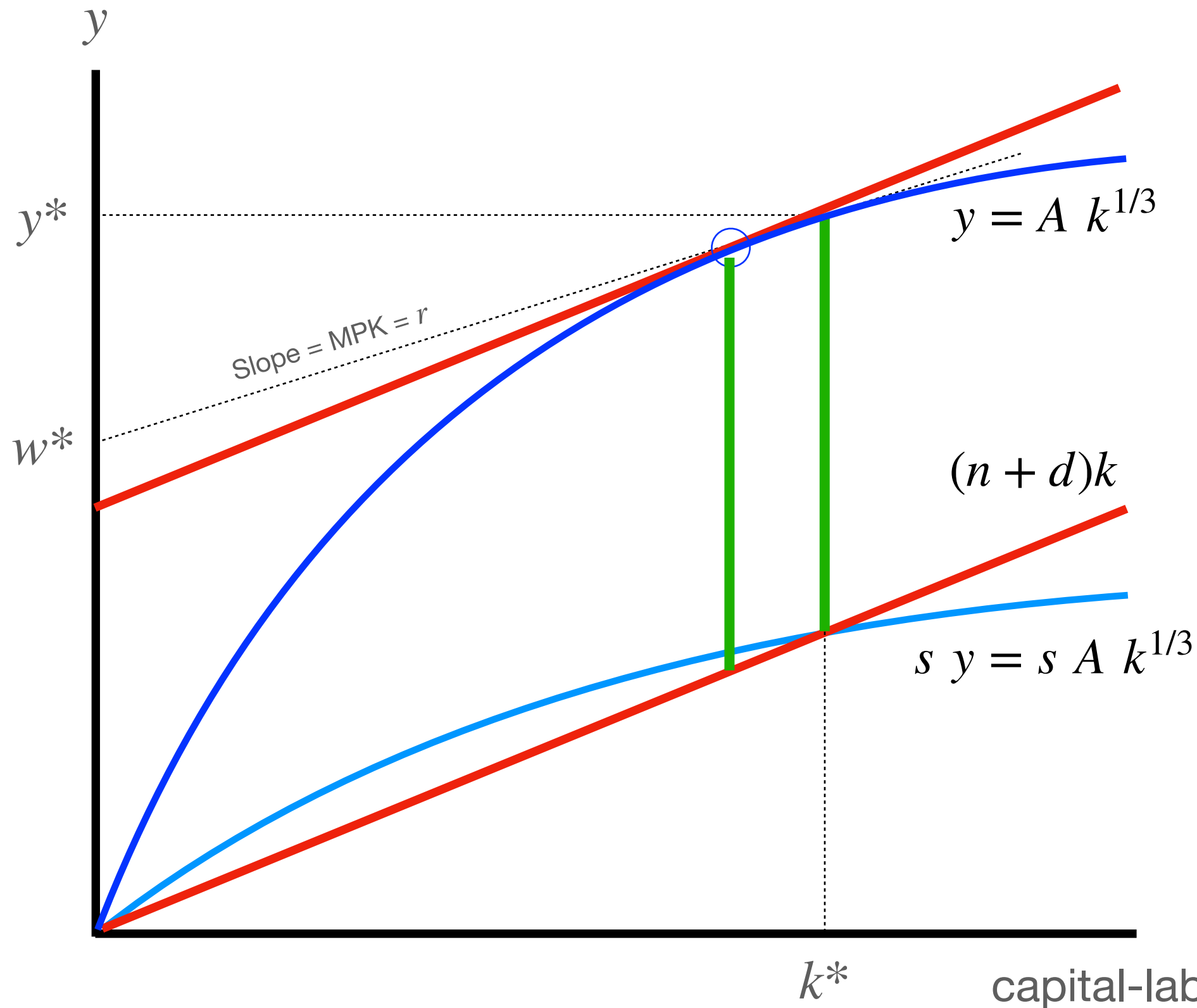
Definitely not here



Also not here



What about here?



- I copied the red depreciation line and pasted a new one tangent to output per worker

- It turns out that is where the green line consumption is longest:

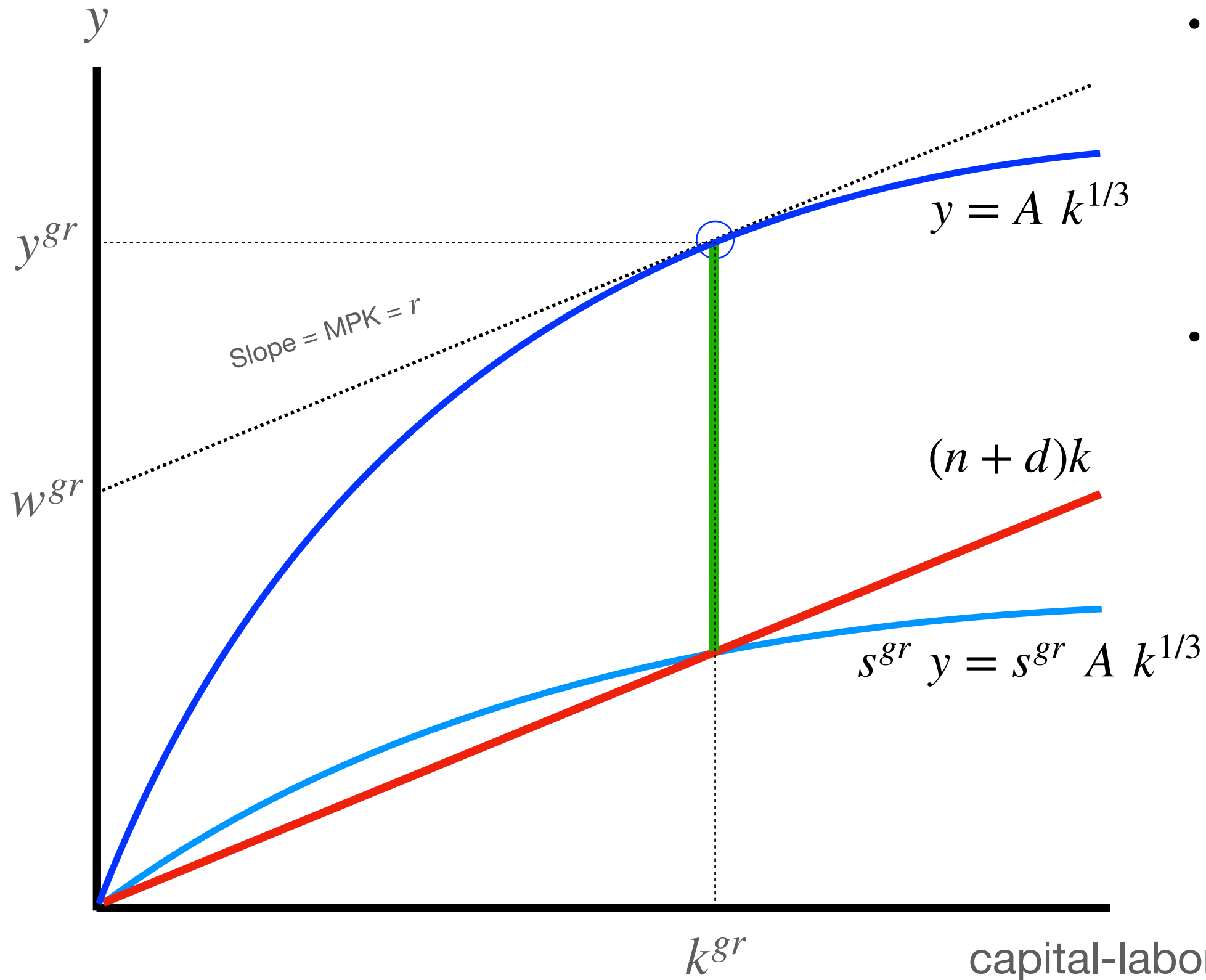
$$c = y - sy$$

$$c = y - (n + d)k$$

- Set $\partial c / \partial k = 0$

$$\partial y / \partial k = MPK = n + d$$

The golden rule rate of saving s^{gr}



- This saving rate maximizes consumption in multiple periods
- The saving rate that maximizes consumption right now only? That is $s^{swindle} = 0$